Synthetic MRI Generation with Latent Diffusion Models

Gian Favero
McGill University
3380 Blvd Robert-Bourassa, Montréal, QC, H2X 2G6
gian.favero@mail.mcgill.ca

Abstract

While deep learning models have shown immense promise and leading performance across a variety of medical imaging tasks related to Multiple Sclerosis (MS), robust and generalizable development is hindered by a barrier to large, unbiased, high-quality, and public MRI data. Generative models such as Diffusion Probabilistic Models (DDPMs) and Generative Adversarial Networks (GANs) provide a solution to this problem by generating realistic synthetic MRI images, however, existing models are yet to strike a balance between computational efficiency, training stability, and image fidelity. In this work, we apply latent diffusion models (LDM) to the task of synthetic MRI generation to show that such a balance can be achieved. We show that LDMs are able to generate high-resolution images that are anatomically plausible, unique enough to introduce diversity to the data, and have pathology consistent with MS. We then compare the performance of our LDM model to the Probabilistic Vision Group's (PVG) state-of-the-art StyleGAN model to show that the LDM is more computationally efficient to train while sacrificing marginal image fidelity in samples. In turn, the effectiveness of LDMs observed in this work demonstrates the transformative potential of computationally efficient diffusion models for high-quality synthetic MRI generation.

1 Introduction

Image-based medicine is of particular use in Multiple Sclerosis (MS), a chronic disease identified by the appearance and enlarging, over time, of T2 lesions in MRI scans of the brain [1]. Deep learning (DL) models in medical image analysis have far outperformed alternative computational and human-based methods in many important contexts such as lesion segmentation [2] and treatment effect estimation [3]. However, the performance of these models is limited by the amount and quality of data available for training, which is often low in medical domains due to difficulties in recruiting subjects and the high cost of acquisition [4]. This issue is further amplified when datasets are held private due to the sensitive nature of personal medical information. Collectively, these factors provide a bottleneck for the development and deployment of robust DL models in image-based medicine and emphasize the need for large, high-quality, bias-free, and public MRI datasets for MS.

Synthetic image generation [5, 6, 7] is a novel approach that helps alleviate this problem by augmenting existing datasets [8]. Generally, this task requires a model to learn how to generate realistic images that are consistent with the underlying data, yet also unique enough to introduce some element of diversity in the data distribution. Successful implementations of generative models can assist in dataset enlargement, debiasing, or even create synthetic public counterparts to private datasets, and in doing so, enable more research to be conducted in image-based medicine for MS. Further, works such as [9] have shown that synthetic images can be used to train downstream models that perform comparably or superior to models trained on real data. Hence, there is real and quantifiable

Submitted to the McGill ECSE 556 course as part of its Final Project. Do not distribute.

value in high-quality synthetic data for the development of robust and generalizable DL models in image-based medicine.

To be deemed useful, a synthetic image generation model must effectively generate convincing representations of real-world entities in high-resolution detail. In the case of brain MRI for image-based medicine research in MS, generated images should be (i) anatomically plausible according to the underlying distribution, (ii) unique and introduce diversity to the data, and (iii) have the pathology consistent with MS (e.g., T2 lesions). In this work, a latent diffusion model (LDM) that satisfies the above criteria is applied to the task of synthetic MRI generation. The contributions of this work are as follows:

1. A novel application of LDMs to the task of 2D synthetic MRI generation for the deployment of more robust and generalizable deep learning models in image-based medicine for MS.
2. A comparison of LDMs to the Probabilistic Vision Group's (PVG) current state-of-the-art StyleGAN model in the context of 2D synthetic MRI generation for MS image-based medicine.

2 Related Work

2.1 Variational Autoencoders

Variational autoencoders (VAE) [6] are a class of generative models that are trained to learn a latent space representation of a data distribution such that the KL-divergence between the latent distribution and a simpler prior distribution (e.g., isotropic Gaussian) is minimized. This latent space can then be randomly sampled and then converted into a realistic image through the network's decoder. VAEs are known to produce blurry images [10] as a result of the learned prior distribution not being representative of the true data distribution. Correspondingly, the image fidelity of VAEs is limited, and for this reason traditional VAEs are not commonly used in the context of MRI generation (though some works, [11], have tried).

2.2 Generative Adversarial Networks

Generative adversarial networks (GAN) [12] are a vastly used class of generative models that have been shown to be effective in generating realistic images in the natural and MRI domain [13]. GANs operate through something analogous to a two-player game in which a "generator" creates images that fool a "discriminator" which is trained to classify images that belong to a ground truth distribution. More formally, the generator, G, maps a random latent vector, z, (sampled from a prior distribution $p(z)$) to an image, x. The discriminator, D, serves as a binary classifier that takes in the generated image, x, as input and outputs a probability that x is a real image. The generator and discriminator are trained simultaneously in an adversarial manner in hopes that a Nash equilibrium is reached and the generator is able to learn the underlying distribution as close as possible. A drawback with such models is that they are known to face challenges such as unstable training, mode collapse, and blurry generation [14]. In the context of MRI generation, GANs have shown a limited ability to operate on full-resolution images due to the high computational cost of training. In turn, scaling GANs to 3D volumes has shown to be too computationally expensive for practical use, drastically narrowing the scope of their application in medical image synthesis.

2.3 Denoising Diffusion Probabilistic Models

Recently, a new class of generative models called diffusion models (DDPM) [7] have been proposed. These models have shown to be superior in training stability, sample fidelity, and sample diversity compared to GANs and VAEs in the natural image domain. DDPMs are based on a stochastic diffusion process that is represented as a Markov chain. Each step in the forward diffusion process, $q(x_t|x_{t-1})$ adds Gaussian noise to an initial image, x_0, from a distribution $q(x)$ over a series of timesteps, T, until it is pure Gaussian noise (x_T). DDPMs learn the reverse Markov chain process, $q(x_{t-1}|x_t)$, by approximating a series of denoising autoencoders conditioned on the input image, x, and the time-step, t. During sampling, noise is iteratively subtracted over a series of T time-steps to obtain a denoised image, $\hat{x}_0$ from a noise sample, $x_T \sim \mathcal{N}(0, I)$.

Though promising, DDPMs operate in the pixel-space, and thus require intensive computational requirements for training and utilize a long, iterative sampling procedure. Works such as [15] propose using diffusion models for unconditional MRI generation. However, such training and evaluating such models requires extensive computations in high-dimensional pixel spaces, requiring GPU resources available only to a small portion of the research field. This poses a significant problem, especially in the scaling of such models to 3D. A more recent class of diffusion models called latent diffusion models (LDM) [16] have been brought forth as a computationally efficient alternative that retain the benefits of DDPMs but operate in a latent space.

3 Methodology

LDMs are a variant of diffusion models that operate in a compressed latent space that is perceptually equivalent to the full resolution pixel space (or as close to it as possible). For a modest trade-off in perceptual information, LDMs are able to generate high-resolution images with a much lower computational cost [16] compared to traditional DDPMs.

3.1 Perceptual Compression

The first step in training an LDM is to compress the input image into a latent space. In the case of brain MRI, the autoencoder is trained to compress an image $x \in \mathbb{R}^{H \times W \times C}$ into a latent code $z \in \mathbb{R}^{h \times w \times c}$ where H, W, C are the height, width, and number of channels of the input image and h, w, c are the height, width, and number of channels of the latent code. The encoder $\mathcal{E}$ maps the input image x to a latent code z such that $z = \mathcal{E}(x)$, and in doing so, downsamples x by a factor of $H/h = W/w$. On the reverse end, the decoder $\mathcal{D}$ maps the latent code z to a reconstructed image $\hat{x}$ such that $\hat{x} = \mathcal{D}(z)$, and in doing so, upsamples z by a factor of $h/H = w/W$. The autoencoder is trained to minimize the reconstruction loss between the input image x and the reconstructed image $\hat{x}$, such that $\hat{x} \approx x$.

The perceptual compression model used in this work is based on [17] — an autoencoder network trained via an objective function consisting of a perceptual loss [18], an adversarial loss [19], an L1 loss, and a slight KL-penalty [6] to enforce the learned latent space toward a Gaussian distribution. The combination of these losses allows the model to learn a latent space that is perceptually equivalent to the input image and robust to noise. L_{AE}, the objective function of the autoencoder, is defined as follows:

$$L_{AE} = \min_{E,D} \max_{\psi} \left(\underbrace{L_{rec}(x, \mathcal{D}(\mathcal{E}(x)))}_{\text{Reconstruction loss}} + \underbrace{L_{adv}(\mathcal{D}(\mathcal{E}(x)))}_{\text{Adversarial loss}} + \underbrace{\log D_{\psi}(x)}_{\text{Patch Discriminator}} + \underbrace{L_{KL}(x; \mathcal{E}, \mathcal{D})}_{\text{KL divergence}} \right) \quad (1)$$

Intuitively, less aggressive downsampling results in a latent code that is more perceptually similar to the input image. Hence, the ultimate objective of the autoencoder is to achieve a balance between the perceptual similarity of the latent code to the input image and the compression of the latent code, such that the diffusion model can be trained efficiently.

There exists various implementations of perceptual compression models in the literature. In this work, we compare the performance of an autoencoder trained from scratch on our dataset to a pre-trained autoencoder from the SDXL [20] release. The SDXL-VAE compresses RGB images of size 256 $\times$ 256 to a four-channel latent code of size 64 $\times$ 64. In this work, we attempt to compress the input images to a three-channel latent code of size 64 $\times$ 64 to show that a more efficient latent space can be learned by training an autoencoder from scratch on our dataset.

We implement the autoencoder from scratch using the `AutoencoderKL` class from MONAI [17], which offers variation in latent code size, number of channels, number of layers, residual blocks, and attention. We base our initial architecture on the model used in [17], which was shown to be a valuable modular component of LDMs in the context of generating medical images from the UK Biobank dataset [21].

3.2 Diffusion Probabilistic Process

Diffusion models [7], which were briefly touched on in Section 2, learn a data distribution $p(x)$ by iteratively denoising a sample generated from pure Gaussian noise. This is equivalent to learning the reverse process of a fixed Markov chain of length T that is initialized with a sample from $p(x)$.

Let $q(x)$ be a real data distribution from which we sample a real image, $x_0 \sim q(x)$. A forward process is defined as $q(x_t|x_{t-1})$ which adds Gaussian noise at each time-step according to a pre-defined noise schedule $0 \leq \beta_1 \leq \beta_2 \leq ... \leq \beta_T \leq 1$. Thus,

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t}x_{t-1}, \beta_t I) \tag{2}$$

Since the forward process is a repeated summation of Gaussian noise, a sample at time-step t can be written as a conditional distribution of x_0, since the sum of Gaussians is also Gaussian. This is a simple property that allows us to add any scaled Gaussian noise to $\mathbf{x}_0$ and end up with x_t for any t:

$$x_t = \sqrt{\overline{\alpha}_t}x_0 + \sqrt{1-\overline{\alpha}_t}\epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

where $\alpha_t = 1 - \beta_t$ and $\overline{\alpha}_t = \prod_{s=1}^{t} \alpha_s$.

The reverse process $q(x_{t-1}|x_t)$ is approximated by a learned function $p_\theta(x_{t-1}|x_t)$ which relies on optimizing the variational lower bound (ELBO) between the true posterior data distribution $q(x_{t-1}|x_t)$ and the learned distribution $p_\theta(x_{t-1}|x_t)$. The objective function can be simplified to the MSE loss on the noise added at time t:

$$L_{DM} = \mathbb{E}_{x,\epsilon\sim\mathcal{N}(0,1),t}\big[\| \epsilon - \epsilon_\theta(x_t, t) \|_2^2 \big] \tag{3}$$

The simplified objective function has been found to be mathematically equivalent to score-matching diffusion models [22], which prove the theory behind diffusion models through annealed Langevin dynamics. Sampling from the diffusion model begins with x_T, an image sampled from a pure Gaussian noise distribution. The reverse process is then run for T time-steps, such that:

$$x_{t-1} \leftarrow x_t - \epsilon_\theta(x_t, t) \tag{4}$$

Eventually, the reverse process converges to a sample from the learned manifold, $\hat{x}_0$. In the original diffusion model [7], stochastic differential equations are used to model the reverse process, requiring the same number of steps as the forward process. However DDIMs [22], have shown that small sacrifices in image fidelity can be had to achieve drastic improvements in sampling efficiency ($10\times$ to $50\times$) by using a deterministic reverse path based on ordinary differential equations.

3.3 Latent Diffusion

The pairing of a perceptual compression model and a diffusion model results in an LDM [16] that is suitable for high-fidelity, computationally efficient MRI image generation. The learned encoder, $\mathcal{E}$, provides access to a lower-dimensional representation of the input images allowing for the forward and reverse diffusion processes to be applied in the latent space, z, which requires less resources than operating in the pixel space. In this manner, the benefits of DDPMs are retained while the computational cost of training is drastically reduced, revealing an avenue to a more optimal balance between computational efficiency, training stability, and image fidelity over DDPMs and GANs.

The underlying engine behind learning the reverse diffusion process is a 2D UNet [23] that is conditioned on the latent code z and the time-step t. The UNet is trained to denoise the latent code z at each time-step t such that the output of the UNet is a denoised latent code $\hat{z}_t$ through the following objective function (adapted from the original DDPM objective function [7]):

$$L_{LDM} = \mathbb{E}_{z,\epsilon\sim\mathcal{N}(0,1),t}\big[\| \epsilon - \epsilon_\theta(z_t, t) \|_2^2 \big] \tag{5}$$

Sampling from the LDM is done by sampling a latent code $z \sim \mathcal{N}(0, 1)$ and passing it through the UNet at each time-step $t \in \{T, ..., 1\}$ to obtain a denoised latent code $\hat{z}_0$. The final output of the LDM is then converted to the pixel space, $\hat{x}_0 = \mathcal{D}(\hat{z}_0)$, where $\mathcal{D}$ is the decoder of the autoencoder.

Various configurations of the UNet have been proposed for efficient training of LDMs, differing in the number of layers, number of channels, residual blocks, and attention. We base our UNet

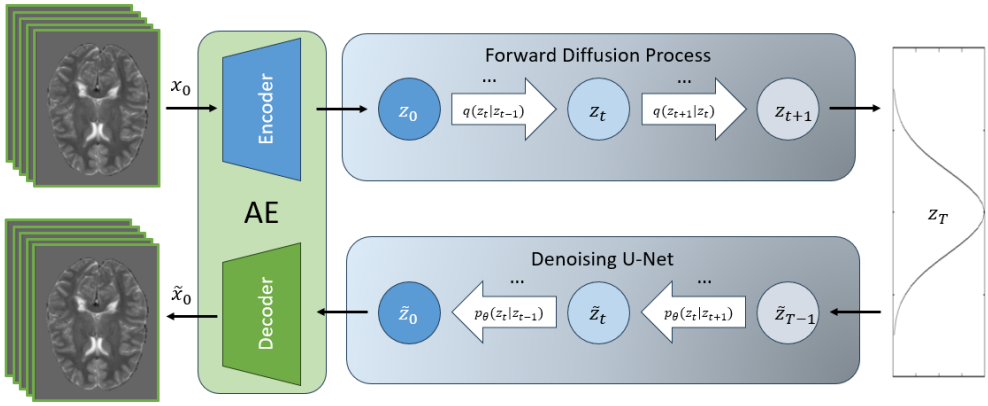


Figure 1: A high-level overview of the LDM architecture. The encoder $\mathcal{E}$ is trained to compress an input image x into a latent code z. The decoder $\mathcal{D}$ is trained to reconstruct the input image x from the latent code z. A forward diffusion process reduces an input latent to pure Gaussian noise, such that a UNet learns how to denoise the latent code z at each time-step t to a latent code $\hat{z}_t$. The final output of the LDM is then converted to the pixel space, $\hat{x}_0 = \mathcal{D}(\hat{z}_0)$.

backbone architecture on the `DiffusionModelUNet` implementation from MONAI [17], which has been shown to be effective with other MRI datasets such as that of the UK Biobank [21]. In terms of noise scheduling, we implement the originally proposed 1000 timestep training schedule with a linear beta schedule using the `DDPMScheduler` class from MONAI [17] for both the forward and reverse diffusion processes.

4 Experiments

4.1 Dataset

The dataset is composed of MRI scans of the brain from ten randomized clinical trials of MS patients: BRAVO [24], ADVANCE [25], ASCEND [26], OLYMPUS [27], CONFIRM ENDORSE [28], DEFINE ENDORSE [28], MAESTRO3, ORATORIO [29], OPERA1[30], and OPERA2 [30], collectively termed LORIS. Each trial enrolled patients with MS. Patient samples consisted of multi-sequence MRI (T1 weighted pre-contrast, T1 weighted post-contrast, FLAIR, T2-weighted, and proton density weighted), lesion maps (T2 hyperintense and gadolinium-enhancing lesions), as well as clinical and demographic features (age, sex, expanded disability status scale scores) at baseline. In total, the dataset consists of $n = 7542$ samples.

This work focuses on the T2-weighted MRI sequences of the samples, which reveal the presence of T2 lesions in the brain indicative of MS [1]. High-resolution MRI images with a resolution of $64 \times 224 \times 176$ voxels at a finer resolution of $1 \times 1 \times 3$ are adapted to 2D axial slices padded to size 256×256 from the central most region of the brain. The images are then skull-stripped and normalized to the range $[0, 1]$.

While MRI images are typically acquired in 3D, the analysis of 2D slices can be advantageous in its effectiveness in reducing computational cost while maintaining information relevant for diagnosis. This work follows such an approach, enabling the use of available computational resources to train a more expressive model in full-resolution of 2D slices, rather than a less expressive model in low-resolution of 3D volumes.

4.2 Evaluation Metrics

4.2.1 Autoencoder

Evaluating an autoencoder is commonly accomplished by measuring the mean absolute error (MAE) and the structural similarity index (SSIM) [31] between the input image and the reconstructed image. The MAE is defined as the average of the absolute difference between the input image and the

reconstructed image, where a lower MAE indicates a better reconstruction. An implementation of L1 loss from PyTorch [32] is used in our experiments.

SSIM is characterized by features of the human visual system such as perceptual quality, luminance, contrast, structure, and scale. MS-SSIM [33] is a variant of SSIM that is robust to structural distortions, aptly used to compare images across multiple resoluions and scales. Both SSIM and MS-SSIM together provide a more holistic evaluation of the perceptual quality of generated or reconstructed images. The metric is bounded between $[0, 1]$ where a higher SSIM indicates more similarity between images. We use the implementation from MONAI [17] to compute the SSIM and MS-SSIM in our experiments.

4.2.2 Latent Diffusion Model

A set of metrics are used to evaluate the computational and time efficiency of training the LDM against an alternative generative model currently used in the PVG. These include the number of steps required to train the model, the wall-clock time required to train the model, and the allowable batch size on a single NVIDIA V100 GPU. This analysis should provide insight into the computational benefits or trade-offs of training an LDM over other types of generative models for the pursuit of higher-fidelity synthetic MRI generation.

To ensure that the LDM is able to generate images that are (i) anatomically plausible and from the underlying distribution, we first observe its image samples qualitatively. Then, we use indirect quantitative metrics like the Frechet Inception Distance (FID) [34]. FID is the current state-of-the-art metric used to evaluate the quality of images created from generative models. It measures the distance between the feature representations of the generated images and the feature representations of the real images, with a lower score indicating more similarity between the distributions. In this work, FID is computed using an InceptionV3 [35] classifier network pre-trained on ImageNet [36]. We generate 1656 synthetic samples to compare against an identical amount of samples from the ground truth distribution. It is important to note that FID is a relative metric in nature, and thus is only useful for comparing the performance of different models on the same dataset for a true evaluation of criteria (i). We additionally consider SSIM and MS-SSIM to assess the anatomical plausibility of the generated images, with the hypothesis that a higher measure indicates a greater indication that criteria (i) is met.

We evaluate (ii) the uniquess of the generated images using the SSIM and MS-SSIM measures between all pairs of generated images to measure the intra-distribution diversity in the synthetic dataset with the expectation that a lower measure indicates more uniqueness and a greater satisfaction of criteria (ii). As with FID, we generate 1656 synthetic samples to compare against an identical amount of samples from the ground truth distribution to compute the SSIM and MS-SSIM scores.

Finally, to ensure that the LDM is able to generate images that have pathology consistent with the underlying distribution, a pre-configured nn-UNet [37] segmentation model is used to generate lesion maps of the generated images. The distribution of lesion count in the generated lesion maps is then compared to the distribution of lesion count in the underlying data to assess the veracity of the generated images in their ability to satisfy criteria (iii). A successful model should generate images with a distribution of lesions that is statistically similar to the distribution of lesions in the underlying data. For this experiment, we generate 500 synthetic samples to compare against the ground truth distribution.

4.3 Computational Resources

All experiments were run on a single NVIDIA V100 GPU with 16 GB of memory. A train-validation-test split of 70-25-5 is performed on the dataset. Given the intensive computational requirements of training generative models no cross validation is performed.

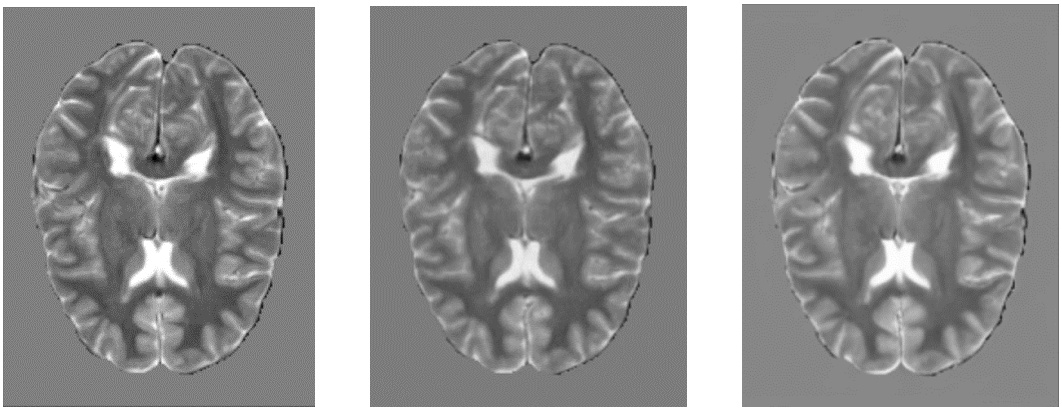
5 Results

5.1 Autoencoder

An implementation of `AutoencoderKL` from MONAI is used to train the autoencoder [17]. A description of the network structure can be found in the Appendix. Following a hyperparameter schedule outlayed in [17], the adversarial, perceptual, and KL losses are weighted by 0.01, 0.001 and

1e-6, respectively. In addition, the patch based discriminator has a warm-up period of 10 epochs prior to being trained. An AdamW optimizer [38] is used to train the discriminator with a learning rate of 5e-4, and the autoencoder with a learning rate of 5e-5. The networks are trained for 116k steps (or 200 epochs), with a batch size for the experiment set to 8 to accommodate the memory constraints of the GPU.

A VAE is additionally considered from the SDXL [20] release, pre-trained (with frozen weights) on a subset of the LAION-5B database [39]. To accommodate the SDXL-VAE model, the T2-weighted channel of each sample is repeated thrice to match the number of input channels of the model. Both autoencoders compress the input images to a 64×64 latent representation, though with three channels for ours compared to four channels for the SDXL-VAE. The autoencoders are evaluated on the validation set in both SSIM and reconstruction loss. The results of the evaluation are shown in Table 1.



(a) Real Image (b) AutoencoderKL (c) SDXL-VAE

Figure 2: Comparison of a real image to reconstructions made by the trained from scratch AutoencoderKL model and pre-trained SDXL-VAE model.

Table 1: Validation Performance Evaluation of Autoencoders

Model	**Latent Dim.**	**SSIM**	**Reconstruction Loss**
AutoencoderKL	3	0.9866	0.0091
SDXL-VAE	4	0.9599	0.0126

The autoencoder trained on our dataset achieves a higher SSIM and lower reconstruction loss than the pre-trained variant. This indicates that MRI images from our dataset belong to a manifold that is not well represented in the LAION-5B subset that the SDXL-VAE was trained on. Hence, clear benefits can be seen in training an autoencoder from scratch on our dataset. As an additional note, our autoencoder achieved this with only three latent dimensions (compared to four for the SDXL-VAE), indicating that the latent space learned by our autoencoder is more efficient than the SDXL-VAE.

5.2 Latent Diffusion Model

We train our LDM model with a 2D UNet backbone based on the `DiffusionModelUNet` implementation from MONAI [17]. We use the trained-from-scratch autoencoder from Section 5.1 as the autoencoder for the LDM. Our LDM based on the SDXL-VAE was unable to converge or produce high-quality samples within 1M training steps (wall-clock equivalent of 5 days).

The LDM is trained for 350k steps with a batch size of 8 and a learning rate of 1e-4. The diffusion process is run for 1000 time-steps with a linear beta schedule. An AdamW optimizer [38] is used to compute and back propagate the training loss. A full description of the network structure and hyperparameters can be found in the Appendix.

We use a StyleGAN [40] implementation, which currently is the highest performing generative model in the PVG, as a baseline for comparison with our LDM. The StyleGAN is trained with its pre-configured hyperparameters on the same data split as the LDM.

5.2.1 Training Comparison

Our LDM model has a clear advantage over the StyleGAN model in terms of training time before convergence. In our experiments, the LDM model converges on its validation loss after 350k steps, which is equivalent to 46.75 hours of training time. In comparison, the StyleGAN model requires 100k steps to converge on its respective validation loss, which is equivalent to 68.25 hours of training time. Using the full GPU memory, the StyleGAN model allows for a greater batch size of 32 compared to 8 for the LDM model, however the benefits of this in terms of training time are offset by the longer training time per step of the StyleGAN model. The training comparison is summarized in Table 2.

Table 2: Training Comparison of LDM and StyleGAN

Model	**Training Steps**	**Wall-Clock Time (hrs)**	**Batch Size**
LDM	350k	46.75	8
StyleGAN	100k	68.25	32

5.2.2 Anatomical Plausibility

A qualitative judgement of the images shown in Figure 3 demonstrates that the LDM model generates realistic images that are anatomically plausible. Both the LDM and StyleGAN models are quantitatively evaluated on their anatomical plausibility of generated images using the Frechet Inception Distance (FID) [34][1].

A synthetic set of images of equal size to the validation set ($n = 1656$) is generated from each model for the quantitative comparisons to take place. Our model achieves an FID score of 1555 while the StyleGAN model achieves an FID score of 1295, indicating that the StyleGAN model generates a data distribution that is marginally closer to the ground truth than our LDM model. In terms of structural similarity, the LDM model achieves an SSIM (MS-SSIM) of 0.7411 (0.7010) between the synthetic dataset and the ground truth validation, substantially higher than the StyleGAN model which achieves an SSIM (MS-SSIM) of 0.3969 (0.4717). This indicates that the samples generated by our LDM model are more similar to the ground truth in terms of details picked up by the structural similarity metrics — some of which are vitally important in the context of medical imaging. Through a combination of qualitative and quantitative approaches, we deem that our LDM satisfies criteria (i). The quantitative comparison is summarized in Table 3.

5.2.3 Sample Diversity

In assessing the diversity of generated samples from our model, a qualitative examination highlights its capacity to produce visually distinct and varied images, as shown in Figure 3. SSIM compliments this analysis with a quantitative metric.

The StyleGAN model achieves an average SSIM (MS-SSIM) of 0.3566 (0.4744) between its generated samples. In comparison, our LDM model achieves an average SSIM (MS-SSIM) of 0.8115 (0.7999) within its synthetic dataset. Evidently, the StyleGAN baseline has greater intra-set diversity in terms of SSIM, however, the similarity indexes indicate that our LDM is producing fairly unique samples, which is in agreement with the qualitative analysis. It is expected that the generated samples have some degree of similarity given that they are all brain MRI images, thus a higher intra-set SSIM score is not entirely indicitive of poor performance, but it could be a result of more consistency in finer details like luminance and scale that is better learned by the LDM than the StyleGAN. To this effect, our LDM satisfies criteria (ii). These quantitative comparisons are also summarized in Table 3.

[1]The FID score is computed using an InceptionV3 [35] classifier network pretrained on ImageNet [36]. Given that medical images are not well represented in ImageNet, a high FID is expected.

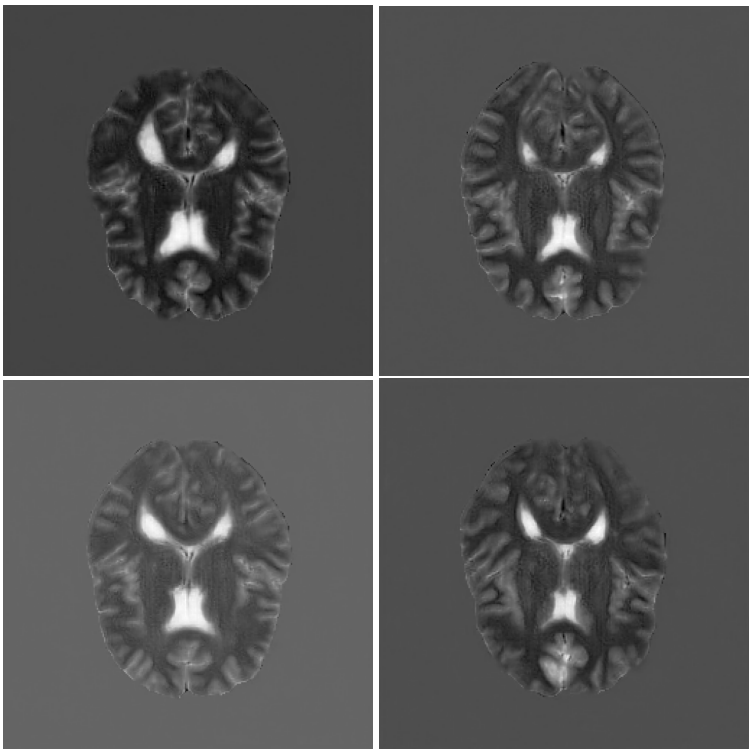


Figure 3: Synthetic brain MRI of MS patients generated by our LDM.

Table 3: Evaluation of Model Performance

Model	FID	SSIM	SSIM (Intra-Set)	MS-SSIM	MS-SSIM (Intra-Set)
LDM	1555	0.7411	0.8115	0.7010	0.7999
StyleGAN	1295	0.3969	0.3566	0.4717	0.4744

5.2.4 Presence of Pathology

We assess the presence of relevant pathology in the synthetic images by generating T2 lesion maps from synthetic images using a pre-configured nn-UNet segmentation model [37] trained on the ground truth images and lesion labels. The distribution of lesion count in the generated lesion maps is then compared to the distribution of lesion count in the ground truth labels to assess if the synthetic images are able to generate images with pathology consistent with its training data. A batch of 500 synthetic images is generated from each of our LDM and the StyleGAN for the quantitative comparisons to take place.

The ground truth distribution (LORIS) has a mean of 10.71 ± 7.31 lesions per sample, while the LDM batch has a mean of 8.45 ± 3.01 lesions per sample, and the StyleGAN has a mean of 14.14 ± 6.52 lesions per sample. Figure 4 shows the density distribution of lesion count in the LORIS, LDM, and StyleGAN datasets. We observe that our LDM model produces a lesion count probability density plot that is significantly more concentrated at the mean than either the ground truth or StyleGAN distributions. However, all three distributions are right-skewed, with a similar mean number of lesions. Hence, the LDM model is able to generate images with pathology consistent with the underlying distribution, satisfying criteria (iii).

6 Conclusion

In this work, we have demonstrated the effectiveness of LDMs in the task of synthetic MRI generation for MS related contexts. Through a multi-faceted, qualitative and quantitative analysis, we have

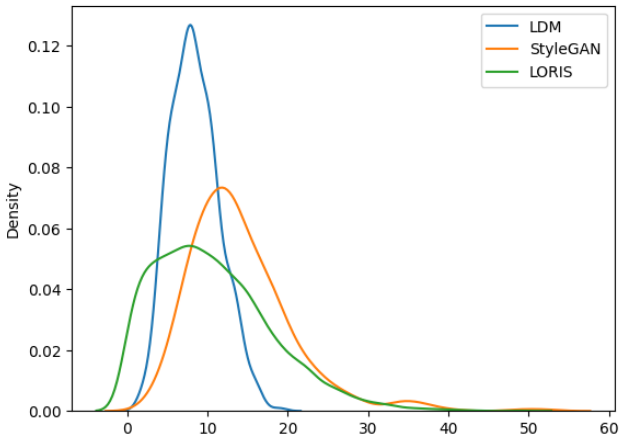


Figure 4: Distribution of lesion count in the LORIS, StyleGAN, and LDM datasets.

shown that LDMs are able to generate high-resolution MRI image samples that are anatomically plausible, unique, and have pathology consistent with a ground truth distribution of MS MRI images. We also compared the performance of our LDM model to the current state-of-the-art generative model in the PVG, a StyleGAN model. Our LDM model was shown to be more computationally efficient than the StyleGAN model in terms of training time per step, trading off slight sacrifices in image fidelity (FID) and intra-set diversity (SSIM and MS-SSIM). With further focus on exploring the use of a more expressive UNet backbone, it is expected that our LDM will outperform the StyleGAN baseline in image fidelity, following similar results in the realm of natural images [16]. This analysis shows that in resource constrained environments, LDMs are a viable alternative to current state-of-the-art generative models for synthetic MRI generation.

Having a DDPM to serve as a baseline would have been advantageous in this work, however, the computational requirements of training a DDPM on our dataset were too intensive to be feasible. In addition, the LDM model was unable to converge when using the pre-trained SDXL-VAE as the autoencoder, indicating that the latent space learned by the SDXL-VAE is not well suited for our dataset. Investigating a fine-tuning approach to the SDXL-VAE may be a fruitful avenue for future work to assess how an autoencoder trained on a much larger dataset can be adapted to our data.

An extension of this work can be made to assess the effectiveness of synthetic data in training downstream models for MS image-based medicine, such as for lesion segmentation models [2], or the study of individual treatment effect estimation [3]. In addition, we will explore the extension of the LDM to MRI sequences beyond T2-weighted images, such as T1-weighted images, as well as scaling upwards to full-resolution 3D MRI images which contain the most relevant information for MS.

The early effectiveness of LDMs observed in this work shows transformative potential of computationally efficient diffusion models for high-quality synthetic MRI generation in image-based medicine.

7 Appendix

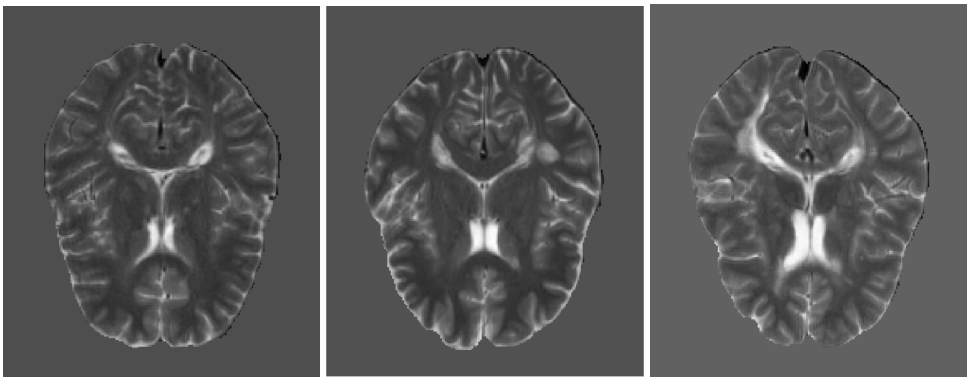


Figure 5: Synthetic brain MRI of MS patients generated by the StyleGAN.

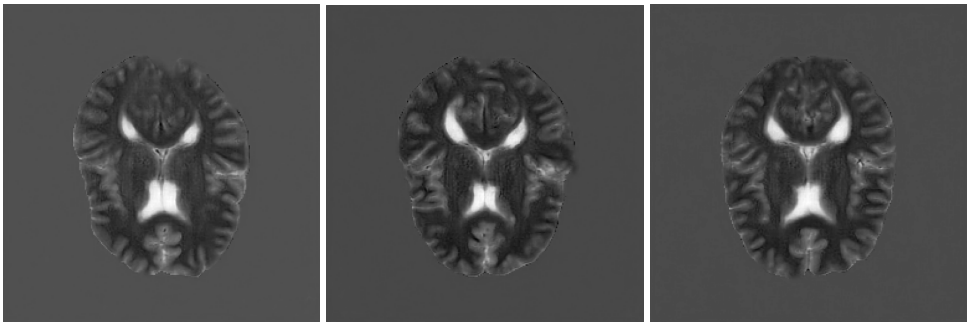


Figure 6: More synthetic brain MRI of MS patients generated by our LDM.

7.1 Autoencoder Hyperparameters

Table 4: AutoencoderKL Hyperparameters

Model Component	**Parameter**	**Value**
AutoencoderKL	Spatial Dimensions	2
	Input Channels	1
	Output Channels	1
	Number of Channels per layer	[128, 128, 256]
	Latent Channels	3
	Number of Residual Blocks	2
	Attention Levels	[False, False, False]
PatchDiscriminator	Spatial Dimensions	2
	Number of Layers (Disc.)	3
	Number of Channels	64
	Input Channels	2
	Output Channels	2
PerceptualLoss	Spatial Dimensions	2
	Network Type	‘alex’
PatchAdversarialLoss	Criterion	‘least_squares’
AdamW	Learning Rate	5e-5
GPU	Batch Size	8
	Num Workers	8

7.2 Latent Diffusion Model Hyperparameters

Table 5: LDM Hyperparameters

Model Component	**Parameter**	**Value**
DiffusionModelUNet	Spatial Dimensions	2
	Input Channels	3
	Output Channels	3
	Number of Residual Blocks	2
	Number of Channels per layer	[128, 256, 512]
	Attention Levels	[False, True, True]
	Number of Head Channels	[0, 256, 512]
DDPM Scheduler	Number of Training Timesteps	1000
	Schedule	Linear Beta
	Beta Start	0.0015
	Beta End	0.0195
AutoencoderKL	Spatial Dimensions	2
	Input Channels	1
	Output Channels	1
	Number of Channels	[128, 128, 256]
	Latent Channels	3
	Number of Residual Blocks	2
	Attention Levels	[False, False, False]
	With Encoder Nonlocal Attention	False
	With Decoder Nonlocal Attention	False
AdamW	Learning Rate	1e-4
GPU	Batch Size	8
	Number of Workers	8

References

[1] Richard A Rudick, Jar-Chi Lee, Jack Simon, and Elizabeth Fisher. Significance of t2 lesions in multiple sclerosis: a 13-year longitudinal study. *Annals of Neurology: Official Journal of the American Neurological Association and the Child Neurology Society*, 60(2):236–242, 2006.

[2] M.M. Weeda, I. Brouwer, M.L. de Vos, M.S. de Vries, F. Barkhof, P.J.W. Pouwels, and H. Vrenken. Comparing lesion segmentation methods in multiple sclerosis: Input from one manually delineated subject is sufficient for accurate lesion segmentation. *NeuroImage: Clinical*, 24:102074, 2019.

[3] Joshua Durso-Finley, Jean-Pierre Falet, Raghav Mehta, Douglas L. Arnold, Nick Pawlowski, and Tal Arbel. Improving image-based precision medicine with uncertainty-aware causal models, 2023.

[4] Hyungjin Chung and Jong Chul Ye. Score-based diffusion models for accelerated mri. *Medical image analysis*, 80:102479, 2022.

[5] Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial networks. *Communications of the ACM*, 63(11):139–144, 2020.

[6] Diederik P Kingma and Max Welling. Auto-encoding variational bayes. *arXiv preprint arXiv:1312.6114*, 2013.

[7] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *Advances in neural information processing systems*, 33:6840–6851, 2020.

[8] Boris van Breugel and Mihaela van der Schaar. Beyond privacy: Navigating the opportunities and challenges of synthetic data. *arXiv preprint arXiv:2304.03722*, 2023.

[9] Shekoofeh Azizi, Simon Kornblith, Chitwan Saharia, Mohammad Norouzi, and David J Fleet. Synthetic data from diffusion models improves imagenet classification. *arXiv preprint arXiv:2304.08466*, 2023.

[10] Huaibo Huang, Ran He, Zhenan Sun, Tieniu Tan, et al. Introvae: Introspective variational autoencoders for photographic image synthesis. *Advances in neural information processing systems*, 31, 2018.

[11] Anna Volokitin, Ertunc Erdil, Neerav Karani, Kerem Can Tezcan, Xiaoran Chen, Luc Van Gool, and Ender Konukoglu. Modelling the distribution of 3d brain mri using a 2d slice vae. In *Medical Image Computing and Computer Assisted Intervention–MICCAI 2020: 23rd International Conference, Lima, Peru, October 4–8, 2020, Proceedings, Part VII 23*, pages 657–666. Springer, 2020.

[12] Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial nets. *Advances in neural information processing systems*, 27, 2014.

[13] Xin Yi, Ekta Walia, and Paul Babyn. Generative adversarial network in medical imaging: A review. *Medical image analysis*, 58:101552, 2019.

[14] Divya Saxena and Jiannong Cao. Generative adversarial networks (gans): Challenges, solutions, and future directions. *ACM Comput. Surv.*, 54(3), may 2021.

[15] Wei Peng, Ehsan Adeli, Qingyu Zhao, and Kilian M Pohl. Generating realistic 3d brain mris using a conditional diffusion probabilistic model. *arXiv preprint arXiv:2212.08034*, 2022.

[16] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10684–10695, 2022.

[17] Walter HL Pinaya, Petru-Daniel Tudosiu, Jessica Dafflon, Pedro F Da Costa, Virginia Fernandez, Parashkev Nachev, Sebastien Ourselin, and M Jorge Cardoso. Brain imaging generation with latent diffusion models. In *MICCAI Workshop on Deep Generative Models*, pages 117–126. Springer, 2022.

[18] Richard Zhang, Phillip Isola, Alexei A. Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 586–595, 2018.

[19] Alireza Makhzani, Jonathon Shlens, Navdeep Jaitly, Ian Goodfellow, and Brendan Frey. Adversarial autoencoders. *arXiv preprint arXiv:1511.05644*, 2015.

[20] Dustin Podell, Zion English, Kyle Lacey, Andreas Blattmann, Tim Dockhorn, Jonas Müller, Joe Penna, and Robin Rombach. Sdxl: Improving latent diffusion models for high-resolution image synthesis. *arXiv preprint arXiv:2307.01952*, 2023.

[21] Cathie Sudlow, John Gallacher, Naomi Allen, Valerie Beral, Paul Burton, John Danesh, Paul Downey, Paul Elliott, Jane Green, Martin Landray, Bette Liu, Paul Matthews, Giok Ong, Jill Pell, Alan Silman, Alan Young, Tim Sprosen, Tim Peakman, and Rory Collins. UK biobank: an open access resource for identifying the causes of a wide range of complex diseases of middle and old age. *PLoS Med*, 12(3):e1001779, March 2015.

[22] Yang Song and Stefano Ermon. Generative modeling by estimating gradients of the data distribution. *Advances in neural information processing systems*, 32, 2019.

[23] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In *Medical Image Computing and Computer-Assisted Intervention–MICCAI 2015: 18th International Conference, Munich, Germany, October 5-9, 2015, Proceedings, Part III 18*, pages 234–241. Springer, 2015.

[24] Channa Kolb-Sobieraj, Sahil Gupta, and Bianca Weinstock-Guttman. Laquinimod therapy in multiple sclerosis: a comprehensive review. *Neurol Ther*, 3(1):29–39, May 2014.

[25] Peter A Calabresi, Bernd C Kieseier, Douglas L Arnold, Laura J Balcer, Alexey Boyko, Jean Pelletier, Shifang Liu, Ying Zhu, Ali Seddighzadeh, Serena Hung, Aaron Deykin, and ADVANCE Study Investigators. Pegylated interferon β-1a for relapsing-remitting multiple sclerosis (ADVANCE): a randomised, phase 3, double-blind study. *Lancet Neurol*, 13(7):657–665, April 2014.

[26] Raju Kapoor, Pei-Ran Ho, Nolan Campbell, Ih Chang, Aaron Deykin, Fiona Forrestal, Nisha Lucas, Bei Yu, Douglas L Arnold, Mark S Freedman, Myla D Goldman, Hans-Peter Hartung, Eva Kubala Havrdová, Douglas Jeffery, Aaron Miller, Finn Sellebjerg, Diego Cadavid, Dan Mikol, Deborah Steiner, and ASCEND investigators. Effect of natalizumab on disease progression in secondary progressive multiple sclerosis (ASCEND): a phase 3, randomised, double-blind, placebo-controlled trial with an open-label extension. *Lancet Neurol*, 17(5):405–415, March 2018.

[27] Kathleen Hawker, Paul O'Connor, Mark S Freedman, Peter A Calabresi, Jack Antel, Jack Simon, Stephen Hauser, Emmanuelle Waubant, Timothy Vollmer, Hillel Panitch, Jiameng Zhang, Peter Chin, Craig H Smith, and OLYMPUS trial group. Rituximab in patients with primary progressive multiple sclerosis: results of a randomized double-blind placebo-controlled multicenter trial. *Ann Neurol*, 66(4):460–471, October 2009.

[28] Ralf Gold, Douglas L Arnold, Amit Bar-Or, Robert J Fox, Ludwig Kappos, Chongshu Chen, Becky Parks, and Catherine Miller. Safety and efficacy of delayed-release dimethyl fumarate in patients with relapsing-remitting multiple sclerosis: 9 years' follow-up of DEFINE, CONFIRM, and ENDORSE. *Ther Adv Neurol Disord*, 13:1756286420915005, May 2020.

[29] Xavier Montalban, Stephen L Hauser, Ludwig Kappos, Douglas L Arnold, Amit Bar-Or, Giancarlo Comi, Jérôme de Seze, Gavin Giovannoni, Hans-Peter Hartung, Bernhard Hemmer, Fred Lublin, Kottil W Rammohan, Krzysztof Selmaj, Anthony Traboulsee, Annette Sauter, Donna Masterman, Paulo Fontoura, Shibeshih Belachew, Hideki Garren, Nicole Mairon, Peter Chin, Jerry S Wolinsky, and ORATORIO Clinical Investigators. Ocrelizumab versus placebo in primary progressive multiple sclerosis. *N Engl J Med*, 376(3):209–220, December 2016.

[30] Stephen L Hauser, Amit Bar-Or, Giancarlo Comi, Gavin Giovannoni, Hans-Peter Hartung, Bernhard Hemmer, Fred Lublin, Xavier Montalban, Kottil W Rammohan, Krzysztof Selmaj, Anthony Traboulsee, Jerry S Wolinsky, Douglas L Arnold, Gaelle Klingelschmitt, Donna Masterman, Paulo Fontoura, Shibeshih Belachew, Peter Chin, Nicole Mairon, Hideki Garren, Ludwig Kappos, and OPERA I and OPERA II Clinical Investigators. Ocrelizumab versus interferon beta-1a in relapsing multiple sclerosis. *N Engl J Med*, 376(3):221–234, December 2016.

[31] Zhou Wang, A.C. Bovik, H.R. Sheikh, and E.P. Simoncelli. Image quality assessment: from error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4):600–612, 2004.

[32] Adam Paszke, Sam Gross, Soumith Chintala, Gregory Chanan, Edward Yang, Zachary DeVito, Zeming Lin, Alban Desmaison, Luca Antiga, and Adam Lerer. Automatic differentiation in pytorch. 2017.

[33] Z. Wang, E.P. Simoncelli, and A.C. Bovik. Multiscale structural similarity for image quality assessment. 2:1398–1402 Vol.2, 2003.

[34] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. Gans trained by a two time-scale update rule converge to a local nash equilibrium. *Advances in neural information processing systems*, 30, 2017.

[35] Christian Szegedy, Vincent Vanhoucke, Sergey Ioffe, Jon Shlens, and Zbigniew Wojna. Rethinking the inception architecture for computer vision. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 2818–2826, 2016.

[36] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E Hinton. Imagenet classification with deep convolutional neural networks. In F. Pereira, C.J. Burges, L. Bottou, and K.Q. Weinberger, editors, *Advances in Neural Information Processing Systems*, volume 25. Curran Associates, Inc., 2012.

[37] Fabian Isensee, Jens Petersen, Andre Klein, David Zimmerer, Paul F Jaeger, Simon Kohl, Jakob Wasserthal, Gregor Koehler, Tobias Norajitra, Sebastian Wirkert, et al. nnu-net: Self-adapting framework for u-net-based medical image segmentation. *arXiv preprint arXiv:1809.10486*, 2018.

[38] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*, 2017.

[39] Christoph Schuhmann, Romain Beaumont, Richard Vencu, Cade Gordon, Ross Wightman, Mehdi Cherti, Theo Coombes, Aarush Katta, Clayton Mullis, Mitchell Wortsman, et al. Laion-5b: An open large-scale dataset for training next generation image-text models. *Advances in Neural Information Processing Systems*, 35:25278–25294, 2022.

[40] Tero Karras, Samuli Laine, and Timo Aila. A style-based generator architecture for generative adversarial networks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 4401–4410, 2019.